

Moussa Atwi

PhD Researcher · Scientific Computing & Machine Learning

Open to industry roles in mathematical modeling, scientific computing, and machine learning.

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 Aachen, Germany

 [Musa Atwi](#)

 [AtwiMusa](#)

 [Portfolio](#)

EXPERIENCE

- PhD Researcher** Full-time · 30 h/wk
RWTH Aachen University · Germany Aug 2023 – Present
"Sparse Exit Wave Reconstruction via Deep Unfolding"
Incorporating data-driven learning into model-based exit wave reconstruction, a variant of phase retrieval, using deep unfolding to create new hybrid approaches.
Phase Retrieval · Image Processing · Optimization · Deep Neural Networks
- Publication**
"A generalization bound for exit wave reconstruction via deep unfolding"
Preprint · related to PhD – [arXiv 2025](#)
- E-Learning & Assessment Assistant** Mini-job · 10 h/wk
RWTH Aachen University · Germany Apr 2026 – Present
Managing exercises and online assessments via Dynexite, publishing materials on Moodle, and handling student grading and feedback.
Dynexite · Moodle · Online Assessment · Grading
- Research Internship & Master's Thesis** Part-time · 20 h/wk
Zuse Institute Berlin · for the M.Sc. at TU Berlin Sep 2022 – May 2023
"Approximation of Koopman Operators Using Random Basis Functions"
Implemented Extended Dynamic Mode Decomposition with random Gaussians and neural networks in Python to investigate Koopman operator theory and nonlinear dynamical systems.
Koopman Operators · Neural Networks · Dynamical Systems
- Research Internship & Master's Thesis** Full-time · 35 h/wk
Université de Bordeaux · for the M.Sc. at UPPA, France Feb – Aug 2020
"Semi-discrete Optimal Transport for Large-Scale Problems"
Implemented gradient descent algorithms in Python to solve large-scale semi-discrete optimal transport problems in image processing and machine learning.
Convex Duality · Image Modeling · Gradient Descent
- Master's Thesis — Control Theory**
Lebanese International University · Lebanon Feb – Aug 2019
"Decay Rates of a Timoshenko System with Kelvin-Voigt Damping"
Proved that the energy of the Timoshenko system with Kelvin-Voigt damping dissipates exponentially or polynomially depending on the material coefficient.
C0-Semigroup · Timoshenko Beam · Polynomial & Exponential Stability
- Mathematics Teacher**
Private Schools · Beirut, Lebanon 2012 – 2018
Worked four years as a school teacher of mathematics in English at private high schools, alongside private tutoring in Real Analysis and Algebra.
Mathematics Teaching · Secondary Education · Private Tutoring

SEMINARS & PROJECTS · FOR THE M.SC. AT TU BERLIN

- Seminar — Optimal Transport**
"Entropic Regularization of Discrete and Continuous Optimal Transport"
Studied Sinkhorn's algorithm, the dual formulation, and convergence under small and large regularization.
Sinkhorn's Algorithm · Dual Formulation · KL-Divergence
- Seminar — Nonlinear Optimization**
"Constrained Optimization and Duality — Applications to ML"
Lagrangian duality and constraint qualifications, formulating the SVM dual, and coordinate and gradient descent methods.
Lagrangian Duality · SVM Dual · Coordinate & Gradient Descent
- Project — Scientific Computing**
"Gas Transport in Porous Medium"
Numerical solution of the nonlinear Porous Medium Equation using the Voronoi Finite Volume Method in Julia, with 1D and 2D simulations.
Porous Medium Equation · Voronoi FVM · Julia

EDUCATION

- M.Sc. Scientific Computing**
Technische Universität Berlin 2021 – 2023
GPA 1.2 · Excellent
- M.Sc. Mathematics, Modeling & Simulation**
Université de Pau et des Pays de l'Adour 2019 – 2020
- M.Sc. Applied Mathematics**
Lebanese International University 2018 – 2019
Rank 1 · GPA 4.0 · Excellent
- Master 1, Pure Mathematics** (1st Year of Master)
Lebanese University 2014 – 2015
- B.Sc. Pure Mathematics**
Lebanese University 2008 – 2012

ACHIEVEMENTS

- Received **A** in the research internship, master's thesis, and most courses of the M.Sc. at TU Berlin.
- Received **A** in all required courses of the M.Sc. at LIU.
- Excellence Scholarship** from Région Nouvelle-Aquitaine to pursue the master's at the University of Pau.
- Ranked 1st** in the Number Theory course, Master 1 at LU.
- Published Arabic poet — **Rashfato Alami**, a collection of 80+ romantic poems.

SKILLS

Scientific & Mathematical

- Optimization
- Machine & Deep Learning
- Deep Unfolding
- Numerical Simulation
- Optimal Transport
- Neural Networks
- Bayesian Inference
- PDEs · FEM / FVM

Computing

- Python
- TensorFlow
- Matlab
- Git
- PyTorch
- Julia
- OpenMP/MPI
- LaTeX

Personal

- Teamwork & Collaboration
- Adaptability
- Problem Solving
- Fast Learner

LICENSES & CERTIFICATES

- Machine Learning Specialization** 2024
Coursera · DeepLearning.AI & Stanford
 - Supervised Learning: Regression & Classification
 - Advanced Learning Algorithms
 - Unsupervised Learning, Recommenders & RL

POETRY

Published Arabic poet — **Rashfato Alami**, a first volume of 80+ romantic poems. Read the collection on the [portfolio](#).

LANGUAGES

Arabic Native **English** C1 **German** B1 **French** A1